Solutions to Problem Set 2

Solution 1. $(a,b)[a,b] = ab$, gives $(a,b) = 20280/780 = 26$. Let $a = 26t, b = 26u$ with $(t,u) = 1$, then $tu = 30$. We now have 4 possible pairs of numbers that multiply 30: $(t,u) = (1,30), (2,15), (3,10), (5,6)$. Thus the solutions are, ignoring the order:

$$(a,b) = (26,780), (52,390), (78,260), (130,156)$$

Solution 2. Let $A(a,b)$, $B(-3,2)$, $C(3,-2)$ be three points in the plane. Then the triangle inequality $AB + AC \geq BC$ gives the required inequality:

$$\sqrt{(a+3)^2 + (b-2)^2} + \sqrt{(a-3)^2 + (b+2)^2} \geq \sqrt{(3+3)^2 + (-2-2)^2}$$
$$\sqrt{a^2+b^2+6a-4b+13} + \sqrt{a^2+b^2-6a+4b+13} \geq 2\sqrt{13}$$

with equality when A lies on BC.

Solution 3. By AM–GM we have

$$\alpha abc > a^3 + b^3 + c^3 \geq 3abc \Rightarrow \alpha > 3$$

To further bound the values of α we argue by contradiction. Supposing $c \geq a+b$, set $c = a + b + x$ for some $x \geq 0$. The condition $a^3 + b^3 + c^3 < \alpha abc$ becomes

$$a^3 + b^3 + (a+b+x)^3 < \alpha ab(a+b+x),$$

or equivalently,

$$x^3 + 3x^2(a+b) + 3x(a^2+b^2) + abx(6-\alpha) + (a+b)\Big(2a^2 + (1-\alpha)ab + 2b^2\Big) < 0.$$

Now we have that

$$2a^2 + (1-\alpha)ab + 2b^2 \geq 0$$

when its discriminant

$$\Delta = \Big((1-\alpha)^2 - 16\Big)b^2 \leq 0$$

which occurs when $-3 \leq \alpha \leq 5$. So, for $3 < \alpha \leq 5$ we have

$$x^3 + 3x^2(a+b) + 3x(a^2+b^2) + abx(6-\alpha) > 0.$$

From the preceding follows that

$$x^3 + 3x^2(a+b) + 3x(a^2+b^2) + abx(6-\alpha) + (a+b)\Big(2a^2 + (1-\alpha)ab + 2b^2\Big) > 0.$$

Contradiction. Therefore, a, b, c are the lengths of the sides of a triangle. We conclude that for $3 < \alpha \leq 5$ the statement holds and we are done.

Solution 4. Take $x = a, y = b, z = ab$ for some a, b integer values. Now:

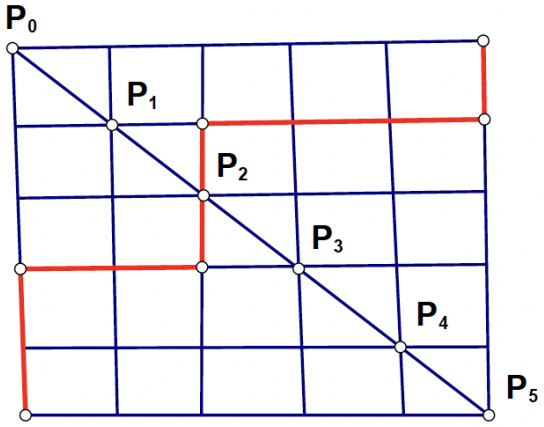
$$\begin{aligned}(x^2+y^2)(x+y)^2+z^2 &= (a^2+b^2)(a+b)^2+a^2b^2\\ &= (a^2+b^2)(a^2+2ab+b^2)+a^2b^2\\ &= a^4+a^2b^2+b^4+2(a^3b+a^2b^2+b^3a)\\ &= (a^2+ab+b^2)^2\end{aligned}$$

Then $t = a^2 + ab + b^2$ gives infinite integer solutions.

Solution 5. (1) Every path must go up m times and to the right n times. In total there must be $m + n$ steps. Consider an alphabet made only of the letters U and R. It is clear that for every path there is one and only one word of $m + n$ letters of this alphabet, where U means to go up and R means to go to the right. For example, in the path of the figure the word would be

$$(U, U, R, R, U, U, R, R, R, U).$$

To form a word with these properties, we only have to choose which are the m places of the $m + n$ possible ones in which we are going to write the letter U. This can be done in $\binom{m+n}{m}$ ways.



(2) Consider an $n \times n$ board divided into unit squares. Then $\binom{2n}{n}$ is the number of paths that move on the sides of the squares, move always up or to the right and go from the bottom left corner to the top right one. Let $P_0, P_1, \ldots, P_n$ be the points in the diagonal of the board that goes through the top left corner. If we want to go from the bottom left corner to the point P_k we have to move k times to the right and $n - k$ times up. There are $\binom{n}{n-k}$ ways to reach this point. To go from P_k to the top right corner of the board, we have to move $n - k$ times to the right and k times up. There are $\binom{n}{k}$ ways of doing this. Thus the number of paths that go through P_k is

$$\binom{n}{n-k}\binom{n}{k} = \binom{n}{k}^2$$

Since every path must go through exactly one point of that diagonal, using the law of the sum n times we obtain

$$\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}$$

Another way to obtain the same results is by counting certain sets. Suppose we have a set of n blue balls and n red balls (all balls can be distinguished from each other). We want to count the number of subsets of n balls. We know there are $\binom{2n}{n}$ such subsets. Let us count how many sets there are such that k of the balls are red. For this we have to choose k red balls and $n-k$ blue ones, which gives

$$\binom{n}{n-k}\binom{n}{k} = \binom{n}{k}^2$$

such sets. Since the number of red balls can vary from 0 to n, we get the desired result.

Solution 6. Since α is an integer and $a_1, a_2, \ldots, a_{2n}$ are distinct integers, then $\alpha - a_1, \alpha - a_2, \ldots, \alpha - a_{2n}$, are also distinct integers such that they satisfy $(\alpha - a_1)(\alpha - a_2)\ldots(\alpha - a_{2n}) = (-1)^n(n!)^2$. That is, $(-1)^n(n!)^2$ is the product of $2n$ distinct integers. On the other hand,

$$(-1)^n(n!)^2 = (-1) \times (1) \times (-2) \times (2) \times (-3) \times (3) \times \ldots (-n) \times (n)$$

is a representation as a product of $2n$ distinct integers which uses all nonzero integers from $-n$ to n inclusive. Now we show that the terms $\alpha - a_i$ are not bigger than n. Indeed, if we suppose that any $|\alpha - a_j| > n$ then

$$|(\alpha - a_1)(\alpha - a_2)\ldots(\alpha - a_{2n})| > 1 \cdot 1 \cdot 2 \cdot 2 \ldots n \cdot n = (n!)^2$$

Contradiction. Hence, $\{\alpha - a_j \mid 1 \leq j \leq 2n\}$ is a rearrangement of the set $\{-i, i \mid 1 \leq i \leq n\}$. So,

$$\sum_{j=1}^{2n}(\alpha - a_j) = \sum_{i=1}^{n}[i + (-i)] = 0$$

from which follows $2n\alpha = a_1 + a_2 + \ldots + a_{2n}$ and $\alpha = \frac{a_1+a_2+\ldots+a_{2n}}{2n}$.